

Marketing Funnel & Conversion Performance Analysis

E-Commerce User Behavior — Funnel & Conversion Report

Data Science & Analytics Internship — Task 3 Submission

Dataset: E-Commerce User Behavior Dataset (Kaggle) | Tool: Python (Jupyter Notebook)

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Executive Summary

This report analyzes the marketing funnel and conversion performance of an e-commerce platform using 500,000 behavioral events from October 2019. The dataset captures every user action across three funnel stages — viewing products, adding to cart, and purchasing — providing a complete picture of where users engage, where they drop off, and how revenue is generated.

The overall funnel converts 8.26% of users from browsing to buying, meaning 91.74% of users who view products leave without purchasing. The most striking finding is that 62.6% of buyers skip the cart entirely and purchase directly — suggesting this platform's checkout experience or one-click purchase path is more commonly used than the traditional cart flow. Electronics dominates both traffic and conversion, Samsung and Apple lead purchase volume, and peak user activity occurs between hours 7 and 9 UTC. Total revenue from this 500,000-event sample reached \$3,115,564 at an average order value of \$319.28.

About This Analysis

Objective: to analyze marketing and lead funnel data to understand how users move from visitors to customers, identify drop-off points, compare conversion performance across product categories, and provide actionable recommendations to improve conversion rates.

- Dataset: E-Commerce User Behavior Dataset (Kaggle) — 500,000 events sampled from October 2019, with 9 fields: event_time, event_type (view/cart/purchase), product_id, category_id, category_code, brand, price, user_id, user_session.
- Tool used: Python 3 in Jupyter Notebook — pandas for data manipulation, matplotlib and seaborn for static charts, plotly for the interactive funnel chart.
- Libraries: pandas, matplotlib, seaborn, plotly, kaleido.
- Data cleaning: converted event_time to datetime; extracted hour, date, and day_of_week features; filled missing brand and category_code with 'Unknown'; extracted main_category from hierarchical category codes (e.g. 'electronics.smartphone' → 'electronics'); dropped rows with null user_id or event_type.
- Funnel approach: counted unique users at each stage rather than raw event counts, since one user can trigger many view events. Purchase path analysis compared users who bought via cart vs. users who purchased directly without ever adding to cart.

Key Performance Indicators

TOTAL USERS 89,124	TOTAL PURCHASES 9,758	OVERALL CONVERSION 8.26%	TOTAL REVENUE \$3,115,564
AVG ORDER VALUE \$319.28	MEDIAN ORDER \$181.53	USERS WHO VIEWED 89,108	DROP-OFF RATE 91.74%

The Marketing Fun



Figure 1. Marketing funnel showing unique users at each stage: Total Users → Viewed → Cart → Purchased.

The funnel reveals a dramatic and characteristic e-commerce drop-off pattern. Almost every user (99.98%) views at least one product, confirming strong traffic arrival. However, only 5% add a product to cart, and 8.26% ultimately purchase. The fact that more users purchase (7,362) than add to cart (4,441) is the defining behavioral pattern of this platform — most buyers skip the cart entirely, a pattern confirmed in the purchase path analysis below.

Funnel Stage	Users	% of Total	Drop-off
Total Users	89,124	100%	—

Viewed Products	89,108	100.0%	0.02%
Added to Cart	4,441	5.0%	95.0%
Purchased	7,362	8.26%	91.74%

Purchase Path: Direct vs Cart

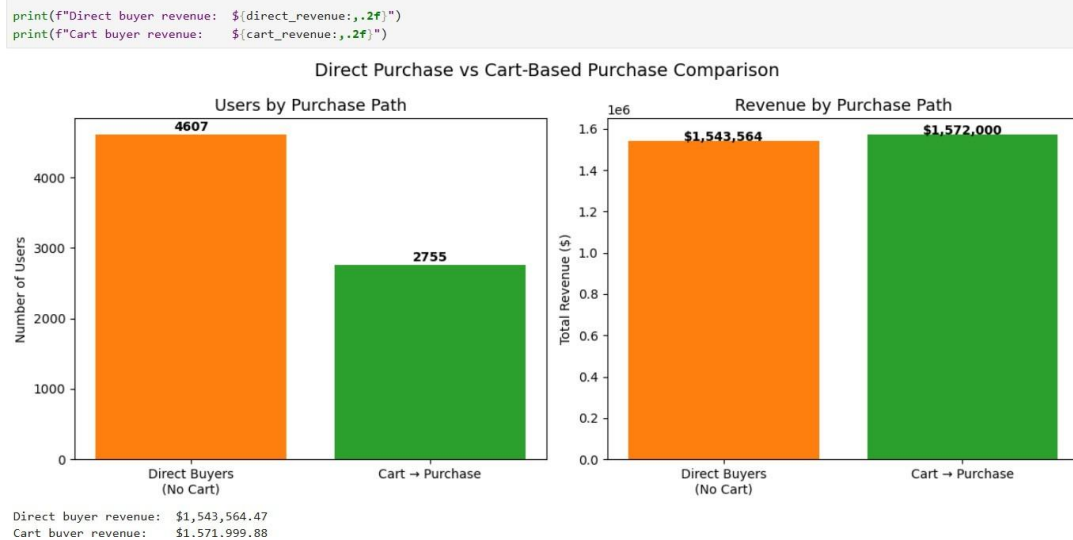


Figure 2. Users and revenue split between direct buyers (no cart) and cart-based buyers.

The most distinctive finding in this dataset is how customers actually buy. Of the 7,362 users who purchased, 4,607 (62.6%) did so directly without ever using the cart, while only 2,755 (37.4%) followed the traditional browse → cart → purchase flow. Despite having fewer users, cart-based buyers generated slightly more revenue (\$1,571,999 vs \$1,543,564), suggesting they buy higher-value items or in larger quantities per session. Direct buyers, though individually generating lower average revenue, represent the majority of the customer base and are the primary revenue driver in terms of volume.

Purchase Path	Users	Revenue	Avg Revenue/User
Direct (No Cart)	4,607 (62.6%)	\$1,543,564	\$335.07
Cart → Purchase	2,755 (37.4%)	\$1,571,999	\$570.60

Conversion Rate by Product Category

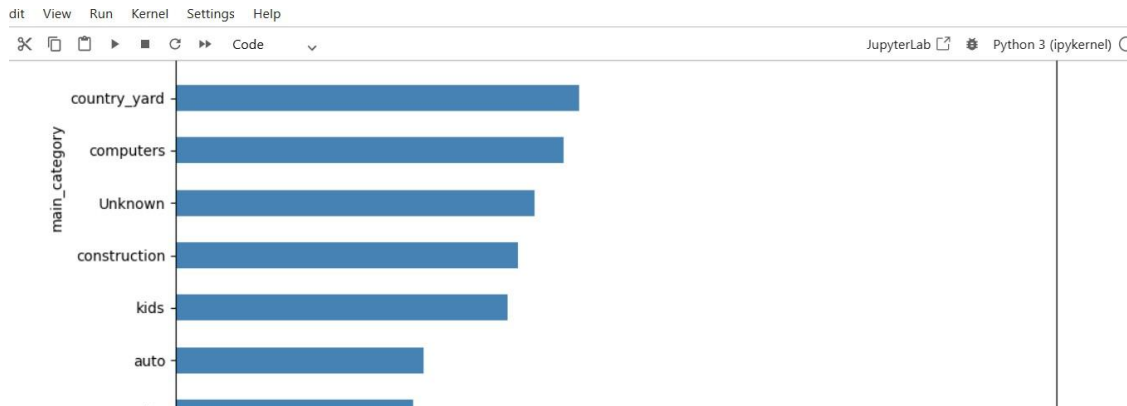


Figure 3. Conversion rate (purchases/views) by main product category, top 10 shown.

Electronics is the standout category — it leads all categories with a 3.18% conversion rate and also has the highest absolute traffic (179,017 views) and purchase volume (5,696 purchases), making it by far the most commercially important category on the platform. Appliances ranks second at 1.73% conversion on 58,296 views, and Computers at 1.47% on 28,177 views. Categories like Auto (0.94%) and Accessories (0.90%) show lower conversion, indicating users browse these categories but rarely commit to purchase — possibly due to higher consideration time or price sensitivity in these segments.

Category	Views	Cart	Purchases	Conv. Rate
electronics	179,017	6,983	5,696	3.18%
appliances	58,296	318	1,010	1.73%
computers	28,177	73	413	1.47%
construction	9,342	85	121	1.30%
kids	6,288	13	79	1.26%
auto	12,367	88	116	0.94%
accessories	2,679	0	24	0.90%

Top Brands by Purchase Volume

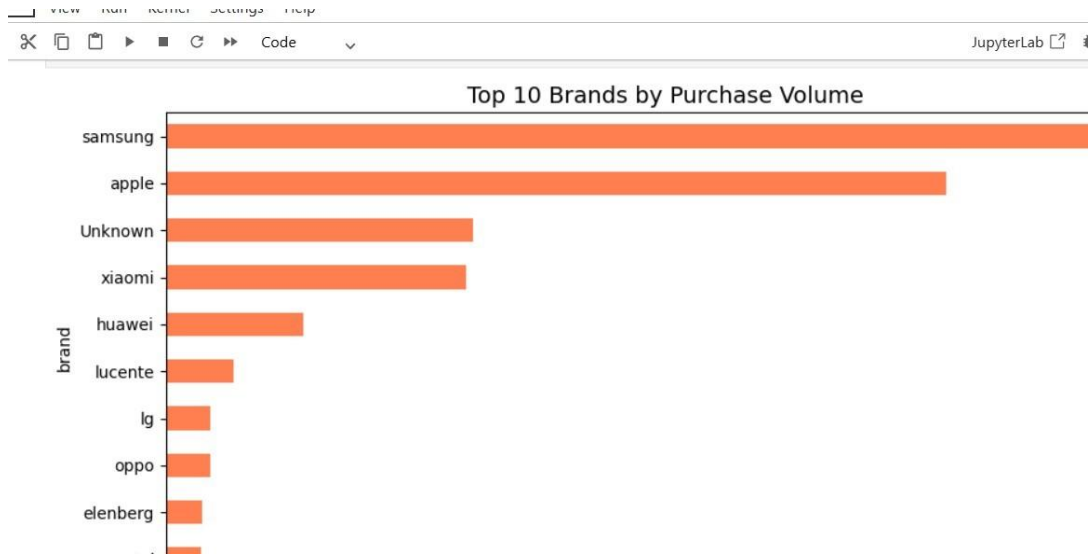


Figure 4. Top 10 brands ranked by number of purchases.

Samsung dominates purchase volume with approximately 2,380 purchases, followed closely by Apple at approximately 1,940 purchases. Together these two brands account for a disproportionate share of all purchases, reflecting the platform's heavy skew toward consumer electronics. Xiaomi and Huawei fill out the top 5, confirming the electronics category's dominance. The large gap between the top 2 brands and the rest of the field suggests the platform's revenue is highly concentrated — protecting and growing the Samsung and Apple product lines should be a top merchandising priority.

Hourly Traffic & Conversion Pattern

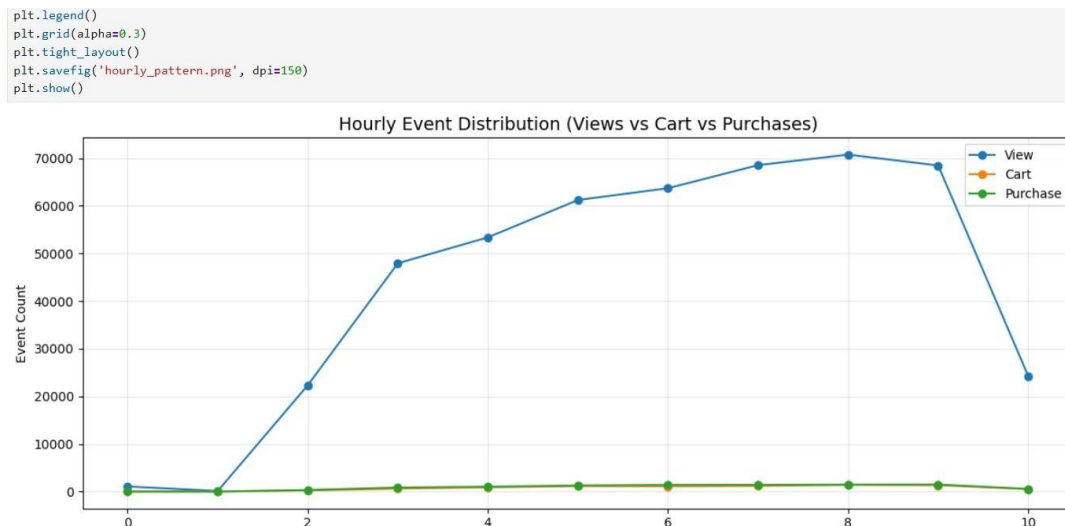


Figure 5. Hourly distribution of views, cart additions, and purchases across the day (UTC).

Traffic builds steadily from near-zero at midnight UTC, rising sharply around hour 2 UTC and reaching a sustained peak between hours 7 and 9 UTC before dropping off sharply at hour 10 UTC. This pattern is consistent with a predominantly European or Indian user base (UTC+1 to UTC+5:30), where hours 7–9 UTC correspond to mid-morning to afternoon local time — the natural peak browsing window for working adults. Cart and purchase events closely mirror the view pattern, confirming that conversion opportunity is highest during peak traffic hours. There is a notable opportunity in the early morning hours (0–1 UTC) where traffic exists but is minimal — targeted campaigns during these off-peak times could capture additional revenue at lower competitive cost.

Key Insights & Recommendations

1. Fix the cart abandonment problem — but reframe it first. The 95% View → Cart drop-off looks alarming, but since 62.6% of buyers never use the cart anyway, the real insight is that the cart is not the primary purchase path on this platform. Focus should shift to optimizing the direct purchase experience (product page CTAs, one-click checkout, trust signals) rather than just pushing users into the cart.
2. Convert the 81,746 browse-only users. Over 91% of users view products but don't buy. Re-engagement campaigns — retargeting ads, email remarketing, push notifications for users who viewed products — are the highest-volume opportunity. Even converting 1% of these non-buyers would add over 800 new purchasers.
3. Double down on Electronics. With a 3.18% conversion rate, 179,017 monthly views, and two dominant brands (Samsung, Apple), electronics is the engine of this business. Invest in better product photography, comparison tools, and faster checkout specifically for electronics. Any friction reduction in this category has outsized revenue impact.
4. Improve Appliances and Computers conversion. Appliances (1.73%) and Computers (1.47%) have solid traffic and reasonable conversion but room to grow. These are highconsideration categories where customers research extensively — adding detailed spec comparisons, customer reviews, and financing/EMI options could meaningfully lift conversion in both segments.
5. Target peak hours with promotions. Hours 7–9 UTC see the highest traffic. Flash sales, limited-time offers, or exclusive discounts deployed during this window would reach the maximum possible audience. Similarly, push notifications or email campaigns sent 1–2 hours before peak (hours 5–6 UTC) would arrive just as users are entering their active browsing period.
6. Increase average order value through cart-based incentives. Cart buyers generate \$570.60 revenue per user vs \$335.07 for direct buyers — a 70% premium. Encouraging more users to use the cart (through bundle deals, 'add one more for free shipping' prompts, or 'customers also bought' recommendations) could significantly lift revenue per transaction even without increasing the number of buyers.

Tools & Deliverables

- Analysis tool: Python 3 (Jupyter Notebook) — pandas, matplotlib, seaborn, plotly.
- Dataset: E-Commerce User Behavior Dataset (Kaggle) — 500,000 events, October 2019.

- Deliverable 1: this report (insights and recommendations).
- Deliverable 2: the Jupyter Notebook (Marketing_Funnel_Analysis.ipynb) with all code, charts, and markdown insights.
- Deliverable 3: a GitHub repository with README, the .ipynb file, and saved chart images.